Diffusion Distillation for Remote Sensing Salient Object Detection

Ziran Gao and Huilan Luo

***Abstract*—Salient object detection (SOD) in optical remote sensing images (ORSIs) is challenging due to limited spatial details, large-scale variation, and cluttered backgrounds with strong look-alike interference. Although diffusion models can progressively refine object structures and boundaries, their multistep sampling introduces high latency, which limits practical deployment over large areas. This article proposes semantic-guided diffusion distillation network (SGDDiff), which distills a high-capacity *feature-space* diffusion teacher into a compact student, thereby retaining diffusion-style refinement while enabling efficient inference. The student is trained under the supervision of a training-only pixel-semantic guidance (PSG) mechanism and intermediate decoder feature alignment. To strengthen representation with low computational overhead, we further develop a lightweight VD encoder (variational gated recurrent neural network (VGRNN)–DyT encoder), which integrates a VGRNN-based refinement unit with a DyT-enhanced Swin-style context block, where the VGRNN incorporates Vision Mamba-based spatial modeling and is further enhanced by a gated depthwise block (GDB). A two-stage training pipeline is adopted, including encoder–decoder pretraining and feature-aligned diffusion distillation. Extensive experiments on multiple ORSI-SOD benchmarks demonstrate that SGDDiff achieves state-of-the-art performance among 18 competing methods while substantially reducing computational overhead, yielding a favorable accuracy–efficiency tradeoff. The code and models are available at https://github.com/zizizirrr/SGDDIff**

***Index Terms*—Dynamic token normalization, generative diffusion, hybrid encoder, knowledge distillation, optical remote sensing, salient object detection (SOD), Vision Mamba.**

I. Introduction

SALIENT object detection (SOD) aims to localize and segment the most visually prominent foreground regions in an image [1]. It serves as a fundamental component for many vision applications, such as image editing, object retrieval, video summarization, weakly supervised learning, and general object detection and segmentation [2], [3]. With the advances of deep neural networks (e.g., convolutional neural networks (CNNs) and vision Transformers), modern SOD approaches have achieved notable progress through multiscale feature aggregation, global context modeling, and boundary-aware refinement [4], [5].

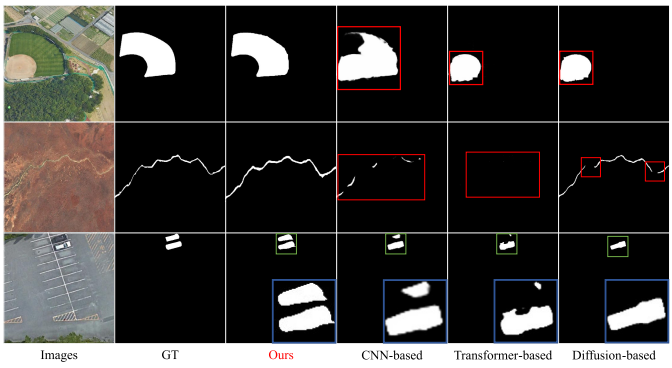


Fig. 1. Limitations of existing methods under typical ORSI-SOD challenges. Qualitative comparison of a CNN-based method (AESINet [7]), a Transformer-based method (MTPNet [8]), a diffusion-based method (ORSIDiff [9]), and the proposed SGDDiff on three typical ORSI-SOD challenges: look-alike clutter (Rows 1 and 2), thin boundaries (Row 3), and small objects (Row 4). Representative error-prone regions are marked by colored boxes, and local zoomed-in views for the small-object case are provided for detailed comparison.

Yet, optical remote sensing image (ORSI)-SOD remains far from solved. In operational remote sensing pipelines, ORSI-SOD is typically performed on *large-area* imagery (e.g., gigapixel scenes processed by tiling), where throughput, memory, and deployment costs impose strict efficiency constraints. Meanwhile, ORSIs exhibit limited spatial details, large-scale variation, and cluttered backgrounds with strong textural/semantic resemblance to foreground objects (e.g., roads versus rooftops, ships versus waves, and buildings versus bare land), making foreground–background separation intrinsically ambiguous [6]. In addition, noisy/weak labels and long-tailed distributions further amplify semantic uncertainty and degrade generalization across regions and acquisition conditions. As a result, practical ORSI-SOD requires a model that is *accurate enough to resolve ambiguity* while remaining *efficient enough to run at scale*.

This tension is clearly reflected by what fails in practice. Most existing ORSI-SOD methods are discriminative CNN/Transformer predictors, which are attractive for fast inference. However, as illustrated in Fig. 1, they often struggle in two typical ways: they may output incomplete foreground

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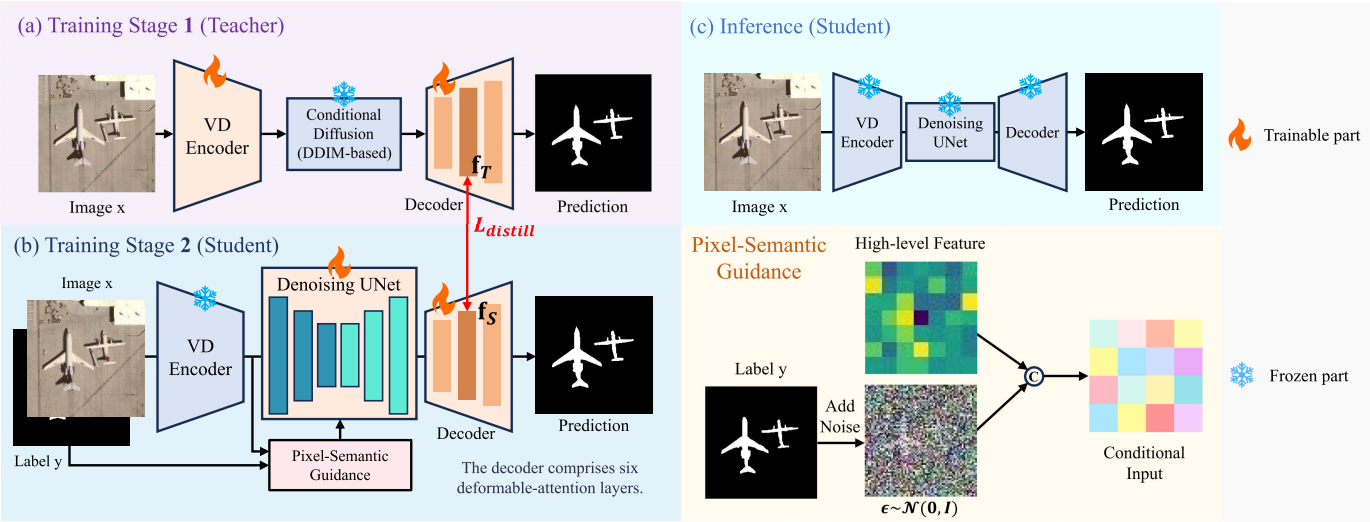


Fig. 2. Overview of the SGDDiff framework. (a) Stage 1: pretrain shared encoder–decoder modules to obtain stable latent representations. (b) Stage 2: distill diffusion knowledge from teacher to student via spatial feature alignment and pixel-level semantic guidance. (c) Inference: the trained student performs efficient saliency prediction without using the teacher.

regions for thin or weak structures, and they may also produce spurious responses in background areas that resemble the salient object. A natural direction is to introduce a stronger refinement mechanism that can progressively correct structure and boundaries. Diffusion models provide such coarse-to-fine refinement via iterative denoising and have shown strong structural coherence in dense prediction [10]. However, Fig. 1 also shows that, under severe look-alike clutter, a traditional diffusion pipeline such as ORSIDiff [9] may still miss weak targets or drift toward background-dominated false activations. Moreover, its long sampling chain incurs prohibitive latency, which conflicts with large-area deployment.

Therefore, Fig. 1 indicates that neither paradigm alone is sufficient for practical ORSI-SOD. Discriminative CNN/Transformer predictors are efficient but tend to be *locally ambiguous and structurally incomplete*, whereas direct diffusion inference is structurally powerful but can be *computationally expensive and semantically fragile* in cluttered ORSI scenes. This motivates a key question: can we retain diffusion-style coarse-to-fine structural refinement, while avoiding heavy sampling at inference and injecting explicit semantics to prevent diffusion collapse under look-alike clutter?

Motivated by this question, we derive four requirements for practical ORSI-SOD: (R1) diffusion-style coarse-to-fine refinement for structurally accurate saliency; (R2) fast inference suitable for large-area processing; (R3) explicit pixel-level semantic guidance to suppress distractors and improve robustness under noisy/weak labels; and (R4) compact yet expressive representations that preserve long-range dependencies and local boundary details under strict computation budgets.

To this end, we propose a semantic-guided diffusion distillation network (SGDDiff) for ORSI-SOD. As shown in Fig. 2, SGDDiff bridges expensive diffusion models and lightweight deployable predictors via a *teacher–student diffusion distillation* paradigm. During training, a high-capacity *feature-space diffusion teacher* denoises and refines encoded image features, transforming noisy latents into structurally coherent representations; a compact student is then trained to mimic this feature refinement with fewer denoising steps. We distill diffusion knowledge through *decoder feature alignment* together with *pixel-level semantic supervision*, enabling the student to produce saliency maps with a coarse-to-fine evolution from global structure to sharp boundaries at substantially reduced inference cost, thereby addressing (R1) and (R2).

To address (R3), we introduce a training-only *pixel-semantic guidance* (PSG) mechanism; ground-truth saliency masks are embedded into a continuous semantic space, perturbed according to the diffusion noise schedule, and fused with encoded visual features to guide the student diffusion model along the denoising trajectory.

To satisfy (R4), we develop a lightweight VD encoder built on the scale-aware modulation Transformer (SMT) [11]. Here, "V" denotes the variational gated recurrent neural network (VGRNN) refinement module and "D" denotes dynamic token normalization (DyT). VD is a *single-stream* hierarchy that fuses a VGRNN cell with a DyT-enhanced Swin-style context block (termed SwinD). The VGRNN injects Vision Mamba-based spatial modeling via a VSB module [12] and is strengthened by our gated depth-wise block (GDB) for lightweight local enhancement and adaptive selection. This fusion yields compact yet expressive features with strong global consistency and boundary sensitivity.

The overall training follows a two-stage strategy (Fig. 2). We first construct a diffusion-augmented teacher representation under frozen diffusion priors and then conduct feature-aligned diffusion distillation, where the teacher guides the student through spatial alignment and semantic conditioning.

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The main contributions of this work are summarized as follows.

1) *SGDDiff for ORSI-SOD:* We propose SGDDiff, a teacher–student diffusion distillation framework that transfers diffusion-style structural refinement from a high-capacity conditional diffusion teacher to a lightweight student, enabling efficient and accurate ORSI-SOD under large-area deployment constraints.
2) *VD Encoder via Single-Stream Architectural Fusion:* We develop a compact VD encoder on top of SMT by integrating VGRNN-based iterative refinement and DyT-enhanced Swin-style context modeling within a unified encoder pipeline, yielding expressive representations that preserve long-range dependencies and fine boundary details in cluttered ORSIs.
3) *Robust Training With Pixel-Level Semantic Guidance and Feature Alignment:* We introduce a pixel-level semantic conditioning strategy with perturbed label embeddings and a two-stage training pipeline (teacher representation construction + feature-aligned distillation) to stabilize optimization and improve robustness to label noise and distribution shifts, achieving a favorable accuracy–efficiency tradeoff on multiple benchmarks.

II. Related Work

A. Salient Object Detection in Optical Remote Sensing Images

Compared with natural-image SOD, ORSI-SOD is more challenging due to large scene coverage, substantial scale variation, small or weak targets, and cluttered backgrounds with high foreground–background similarity. These characteristics require models with stronger multiscale representation, contextual modeling, boundary preservation, and background suppression.

To address these challenges, many ORSI-specific methods have been developed. CNN-based models such as LVNet [13], DAFNet [14], and ORSI [15] mainly improve multiscale feature aggregation and region–boundary modeling. Subsequent methods, including ERPNet [16], AESINet [7], RAGRNet [17], and MEANet [18], further enhance saliency localization by exploiting edge guidance, semantic–boundary interaction, dynamic reasoning, and interactive refinement.

More recently, hybrid and long-range modeling strategies have been introduced to improve robustness in complex scenes. ASNet [19] combines Transformer and CNN representations for adaptive semantic matching and multiscale reasoning, while MTPNet [8] introduces task prompts and task tokens to jointly optimize saliency and edge perception in a lightweight multitask framework.

Meanwhile, efficiency-oriented ORSI-SOD has attracted increasing attention for large-area inference and deployment on resource-constrained platforms. CorrNet [20], SAFINet [5], and LiteSalNet [6] improve the accuracy–efficiency tradeoff through lightweight feature interaction, spatial-attention feedback refinement, and multistream progressive decoding, respectively. Despite these advances, existing lightweight models may still suffer from degraded structural completeness and boundary quality in complex scenes, leaving lightweight yet structurally faithful ORSI-SOD an open challenge.

B. Diffusion Models and Conditional Generation

Diffusion probabilistic models have demonstrated strong capability in image synthesis and dense prediction by modeling complex data distributions through iterative denoising [10], [21]. DDIM [22] further improves sampling efficiency through a non-Markovian formulation. In segmentation, diffusion-related frameworks have explored leveraging generative priors and diffusion-model representations for mask prediction and refinement. For example, DiffSeg [23] exploits self-attention maps from a pretrained stable diffusion model for segmentation, while FocusDiffuser [24] introduces a diffusion-based framework to enhance detail perception in camouflaged object detection. For ORSI-SOD, ORSIDiff [9] formulates saliency detection as conditional diffusion-based mask generation from noise.

Our design differs fundamentally from prior diffusion-based ORSI-SOD methods, especially ORSIDiff [9]. ORSIDiff performs mask-space conditional generation and therefore retains a costly multistep diffusion process during inference. In contrast, SGDDiff employs diffusion as a training-time feature-space teacher, where the teacher refines noised visual features, and the student absorbs diffusion-enhanced structural priors through decoder feature alignment and PSG. As a result, SGDDiff transfers diffusion-enhanced structure and label-driven semantics into a compact predictor, rather than directly relying on iterative sampling at inference.

Despite their strong structural priors, diffusion-based dense prediction methods generally incur high inference cost due to multistep sampling and often require carefully designed conditioning under complex backgrounds. In contrast, our method adopts a feature-aligned diffusion distillation paradigm with pixel-level semantic guidance, enabling efficient transfer of diffusion-style structural refinement into a lightweight student model.

C. Knowledge Distillation and Spatial Alignment

Knowledge distillation is a widely used strategy for compressing dense prediction models while preserving performance [25]. Beyond final-output matching, recent studies highlight the importance of transferring intermediate representations and structured knowledge. For instance, IFVD [26] distills intraclass feature variation through pixel-to-prototype similarity transfer, while MasKD [27] uses learnable pixel-wise masks to emphasize informative regions during feature distillation (FD).

Motivated by these advances, we transfer intermediate decoder features from a diffusion teacher to a lightweight student via spatial feature alignment. Such alignment helps the student approximate the teacher's representation space, which is particularly beneficial for ORSI-SOD, where cluttered backgrounds and large-scale variation often degrade structural integrity and boundary precision.

D. Encoder Optimization and Dynamic Modeling

Accurate ORSI-SOD requires encoders that can capture long-range dependencies while preserving fine-grained local

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structures under large-scale variation and cluttered backgrounds. Existing studies have explored encoder optimization through multiscale interaction, adaptive feature modulation, and boundary-aware enhancement.

Representative efforts include the SMT [11] and DyT [28], which improve scale-aware representation and optimization stability, respectively. Related dynamic modeling ideas have also been explored in low-level vision to improve representation flexibility under complex degradations [29], [30], [31], [32]. In ORSI-SOD, related methods also enhance feature discrimination through interaction modules and edge-aware refinement [6], [7], [33]. However, these methods are mainly designed as discriminative encoders and are not tailored to diffusion-guided representation transfer.

Motivated by these observations, we design a *single-stream hybrid encoder* that integrates efficient sequence modeling and dynamic token calibration. Specifically, the proposed VD encoder combines VGRNN-based iterative refinement, Vision Mamba-driven spatial modeling, and a DyT-enhanced Swin-style context module to learn compact multiscale features. This design is better aligned with the subsequent diffusion distillation stage, helping suppress clutter interference and preserve thin or small salient structures in ORSIs.

III. Methodology

A. Framework Overview

This section outlines the overall pipeline of SGDDiff for ORSI-SOD. As shown in Fig. 2, SGDDiff adopts a two-stage training paradigm—*teacher representation construction* and *feature-aligned diffusion distillation*—and performs efficient inference with a lightweight student denoiser. The key idea is to retain diffusion-style coarse-to-fine refinement while avoiding heavy sampling and model overhead at deployment.

1) Stage 1 (Teacher Representation Construction): We first construct a diffusion-augmented teacher to provide stable structural priors for distillation. The teacher comprises the VD encoder, a pretrained six-layer U-Net diffusion backbone [34], and a multilayer decoder. Here, "feature-space diffusion" means that diffusion is performed on noised encoder features rather than on image pixels or output masks. During this stage, the diffusion backbone is kept frozen and serves as a structural projector that refines encoded visual features into diffusion-enhanced representations. The VD encoder and decoder are trained under saliency supervision to produce semantically discriminative and structurally coherent outputs, which later act as reliable distillation targets.

2) Stage 2 (Feature-Aligned Diffusion Distillation Into a Compact Student: After the teacher converges, we distill its knowledge into a lightweight student that shares the same VD encoder and decoder but replaces the heavy diffusion backbone with a compact denoiser. The distillation is driven by two complementary signals: 1) *decoder feature alignment* to transfer diffusion-enhanced structural representations and 2) a training-only *PSG* mechanism that injects dense pixel-wise semantics along the denoising trajectory to suppress clutter-induced ambiguity and improve robustness.

3) Inference (Efficient Student Prediction): At test time, the teacher and the PSG branch are removed. Only the VD encoder, the lightweight denoiser, and the decoder are retained, enabling fast saliency prediction with a favorable accuracy–efficiency tradeoff for practical large-area ORSI-SOD deployment.

B. Architectural Fusion in the VD Encoder: VGRNN Meets SwinD

ORSI-SOD requires encoders that can capture long-range dependencies to suppress background look-alikes while preserving fine local structures for thin boundaries and small targets under limited computational budgets. To this end, we develop a lightweight hybrid encoder, termed the VD encoder (VGRNN–DyT encoder), based on a four-stage SMT backbone [11]. Our VGRNN refinement module is inserted after Stages 2–4 to iteratively enhance hierarchical features. As shown in Fig. 3(a), VD is a *single-stream* encoder that achieves *architectural fusion* by combining: 1) VGRNN-based refinement, which introduces Vision Mamba-driven spatial modeling into gated recurrent updates for selective long-range propagation; and 2) a DyT-enhanced Swin-style context block, termed SwinD, which strengthens multiscale context aggregation and improves optimization stability under heterogeneous ORSI distributions. This design yields compact yet expressive representations with enhanced semantic discrimination and boundary sensitivity, thereby benefiting subsequent diffusion distillation.

1) VGRNN Cell: Accurate ORSI-SOD relies on global semantic consistency and fine-grained local details (e.g., thin boundaries and texture variations). While Transformer-based encoders are effective for long-range modeling, they may underemphasize subtle local structures when foreground objects are visually similar to cluttered backgrounds. Inspired by VMRNN [12], we develop an improved VGRNN cell [Fig. 3(c)] that couples Vision Mamba-based spatial modeling with gated recurrent refinement. Notably, the recurrence is employed for *iterative feature refinement* rather than temporal sequence modeling.

Let X_t denote the input features at refinement iteration t, and we set the number of iterations to $T = 3$ by default. The VGRNN maintains a hidden state H_t and a memory cell C_t. We first form a joint representation

$$Z_t = \mathrm{LP}\left(\mathrm{Concat}\left(X_t, H_{t-1}\right)\right) \tag{1}$$

where $\mathrm{LP}(\cdot)$ denotes a linear projection. The update is defined as

$$W_t = \mathrm{GDB}\left(Z_t\right) \tag{2}$$

$$F_t = \sigma\left(W_t \odot \mathrm{VSB}\left(Z_t\right)\right) \tag{3}$$

$$C_t = F_t \odot \left(\tanh\left(\mathrm{VSB}\left(Z_t\right)\right) + C_{t-1}\right) \tag{4}$$

$$H_t = F_t \odot \tanh\left(C_t\right) \tag{5}$$

where $\sigma(\cdot)$ is the sigmoid function and $\odot$ denotes the element-wise multiplication. The states are initialized as $H_0 = \mathbf{0}$ and $C_0 = \mathbf{0}$. $\mathrm{VSB}(\cdot)$ is the Vision Mamba-based spatial block [12] [Fig. 3(e)] that injects efficient long-range spatial context,

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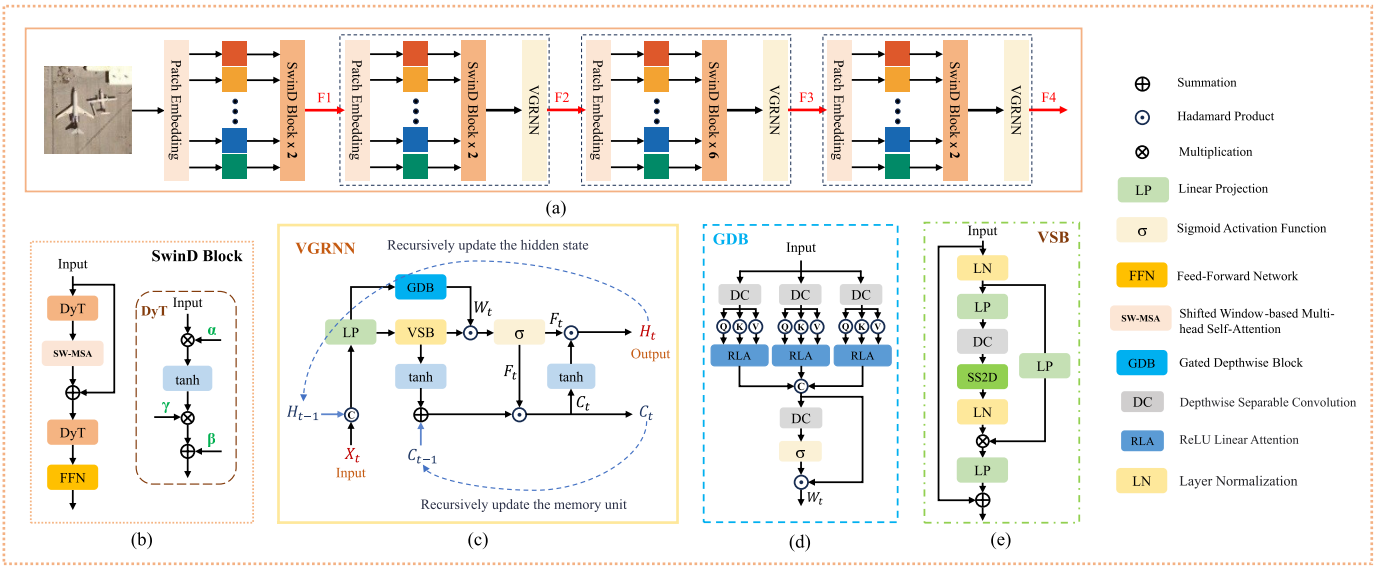


Fig. 3. Proposed VD encoder architecture. (a) Overview. (b) SwinD block: normalization layers are replaced with dynamic token normalization (DyT) for improved stability. (c) VGRNN cell: couple Vision Mamba-based spatial modeling (VSB) with gated recurrent refinement, enhanced by a GDB. (d) GDB block: lightweight local enhancement and adaptive feature selection. (e) VSB block: key Mamba modules for efficient long-range spatial modeling.

while our GDB(·) provides lightweight local enhancement and adaptive feature selection to suppress background noise. After T iterations, the final hidden state H_T is forwarded to the next encoding stage.

2) Gated Depthwise Block: Our GDB [Fig. 3(d)] enhances local structures and selects informative responses for the VGRNN update with minimal overhead. It first enriches local cues via multibranch depthwise separable convolutions and then applies branch-wise ReLU linear attention (RLA) followed by feature fusion and gating.

a) Multibranch depthwise separable convolution: Given an input feature sequence $Z_t \in \mathbb{R}^{N\times C}$, we construct three depthwise separable convolution branches with kernel sizes $(3, 5, 3)$

$$Z_t^{(s)} = \mathrm{DC}_{k_s}(Z_t), \quad k_s \in \{3,5,3\}, \ s \in \{1,2,3\} \tag{6}$$

where $\mathrm{DC}_{k_s}(\cdot)$ denotes a depthwise separable convolution with kernel size $k_s \times k_s$, and each branch has its own parameters. This produces three locally enhanced sequences $\{Z_t^{(s)}\}_{s=1}^{3}$.

b) Branch-wise projection and RLA: For each branch s, we compute

$$Q^{(s)} = Z_t^{(s)} W_Q^{(s)}, \quad K^{(s)} = Z_t^{(s)} W_K^{(s)}, \quad V^{(s)} = Z_t^{(s)} W_V^{(s)} \tag{7}$$

where $W_Q^{(s)}, W_K^{(s)}, W_V^{(s)} \in \mathbb{R}^{C\times d}$ are learnable weights. Following [35], we apply ReLU-based linear attention with $\phi(\cdot) = \mathrm{ReLU}(\cdot)$:

$$O_i^{(s)} = \frac{\phi\left(Q_i^{(s)}\right)\left(\sum_{j=1}^{N}\phi\left(K_j^{(s)}\right)^{\top} V_j^{(s)}\right)}{\phi\left(Q_i^{(s)}\right)\left(\sum_{j=1}^{N}\phi\left(K_j^{(s)}\right)^{\top}\right)}. \tag{8}$$

The two summations are computed once per branch and reused for all queries, yielding $\mathcal{O}(N)$ time and memory complexity.

c) Feature fusion and gating: We concatenate the three branch outputs and generate a spatially aware gate using a 3×3 depthwise separable convolution

$$U_t = \mathrm{Concat}\left(O^{(1)}, O^{(2)}, O^{(3)}\right) \tag{9}$$

$$W_t = \sigma\left(\mathrm{DC}_3\left(U_t\right)\right) \odot U_t \tag{10}$$

where $\sigma(\cdot)$ is the sigmoid function. The gate $\sigma(\mathrm{DC}_3(U_t))$ adaptively amplifies informative responses and suppresses background-dominated activations. The resulting W_t is then fed into the VGRNN cell to modulate the hidden and cell-state updates.

3) SwinD Block: Standard Transformer blocks typically rely on normalization layers (e.g., LayerNorm) to stabilize feature distributions. However, ORSIs exhibit large distribution shifts across regions and acquisition conditions, making training sensitive to feature calibration. To improve stability, we adopt dynamic token normalization (DyT) [28], a learnable element-wise calibration that replaces conventional normalization. When integrated into a Swin-style context modeling block, we refer to the resulting module as SwinD [Fig. 3(b)].

DyT performs a pointwise transformation

$$\mathrm{DyT}(x) = \gamma \cdot \tanh(\alpha x) + \beta \tag{11}$$

where α is a learnable scalar and (γ, β) are affine parameters. The bounded $\tanh(\cdot)$ suppresses extreme activations while maintaining gradient flow, improving convergence and robustness under heterogeneous ORSI distributions. In our implementation, DyT systematically replaces normalization layers within the VD encoder, contributing to more stable training and improved representational quality for ORSI-SOD.

C. PSG Strategy

In conditional diffusion models, generation is usually guided by auxiliary signals such as image features, text prompts,

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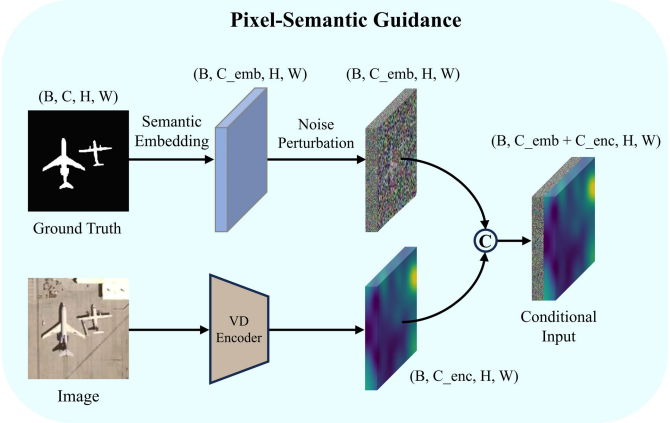


Fig. 4. Illustration of the proposed PSG strategy. During *training*, ground-truth saliency maps are transformed into continuous semantic embeddings, perturbed with Gaussian noise according to the diffusion schedule, and concatenated with visual features from the VD encoder to form the conditional input.

or other high-level priors. For dense prediction tasks (e.g., semantic segmentation and saliency detection), however, the ground-truth masks themselves provide dense, pixel-level semantic supervision that is usually exploited only through loss functions, rather than being explicitly injected into the generative process. In this work, we leverage these masks during training as an additional conditioning signal: by embedding ground-truth labels into a continuous semantic space and feeding their noised versions into the diffusion module, we directly expose the model to the desired pixel-wise semantics along the denoising trajectory. This PSG helps bridge the gap between generic denoising and task-specific saliency prediction, encouraging the network to learn representations that are more aligned with foreground–background separation and more robust to cluttered ORSI scenes.

As illustrated in Fig. 4, the core idea is to embed discrete saliency labels into continuous vectors, perturb these embeddings with Gaussian noise following the diffusion schedule, and fuse the noisy semantic features with image-based encodings. This fused representation is used as an additional training-time condition for the diffusion module; at inference, the semantic branch is removed and the model relies solely on image features.

1) Semantic Embedding Construction: Let $Y \in \mathbb{R}^{H\times W}$ denote the ground-truth saliency mask, where each pixel indicates foreground or background. A learnable embedding layer $E{:}\mathbb{N} \to \mathbb{R}^d$ transforms the mask into a d-dimensional feature map

$$F_{\text{sem}} = E(Y) \in \mathbb{R}^{H\times W\times d}. \tag{12}$$

This preserves the spatial layout of the labels while providing a differentiable per-pixel semantic representation.

2) Semantic Perturbation: To align with the forward noising process in diffusion models, we perturb the semantic embeddings by injecting Gaussian noise at a randomly sampled timestep t

$$y_t = \sqrt{\bar{\alpha}_t}\cdot F_{\text{sem}} + \sqrt{1-\bar{\alpha}_t}\cdot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I) \tag{13}$$

where $\bar{\alpha}_t$ is the cumulative noise coefficient. This transforms the discrete label signal into a smooth, stochastic, and temporally coherent conditioning input for training the diffusion module.

3) Image Feature Extraction: In parallel, the input image $x \in \mathbb{R}^{H\times W\times 3}$ is fed into the VD encoder to obtain visual features

$$z_v = \text{VD}(x) \in \mathbb{R}^{H\times W\times C} \tag{14}$$

which encode global context, texture, and structural cues.

4) Conditional Feature Fusion: During training, the noisy semantic condition y_t and the visual feature map z_v are concatenated along the channel dimension

$$z_c = \text{Concat}(z_v, y_t) \in \mathbb{R}^{H\times W\times (C+d)} \tag{15}$$

and z_c is fed into the conditional diffusion module. This fused representation tightly couples pixel-level semantic priors with image features, guiding the model toward semantically aligned and structurally coherent saliency maps.

D. Two-Stage Training With Decoder FD

To transfer the representational capacity of the diffusion-augmented teacher network into a lightweight student model, we adopt a two-stage training strategy. Stage 1 learns a strong encoder–decoder representation under fixed diffusion priors, and Stage 2 distills this representation into a compact model by aligning decoder feature maps in a spatially consistent manner.

1) Stage 1 (Teacher Semantic Representation Construction): As shown in Fig. 2(a), the teacher network comprises a VD encoder, a pretrained DDIM-based six-layer U-Net sampler [22], and a six-layer deformable-attention decoder [34]. In this stage, the diffusion sampler is kept *frozen* and only the VD encoder and decoder are updated. Given an input image x, the VD encoder produces latent features z_0, which are refined by the fixed diffusion backbone using a 50-step noise schedule and a 10-step DDIM sampler to obtain diffusion-enhanced latents $\tilde{z}_0$. The decoder then maps $\tilde{z}_0$ to a saliency prediction $\hat{Y}_T$.

The teacher decoder is supervised by an SOD loss combining binary cross-entropy (BCE) and Dice loss [36]

$$\mathcal{L}_{\text{task}}^{(\text{teacher})} = \mathcal{L}_{\text{BCE}}(\hat{Y}_T, Y) + \mathcal{L}_{\text{Dice}}(\hat{Y}_T, Y) \tag{16}$$

where $\hat{Y}_T$ and Y denote the predicted and ground-truth saliency maps, respectively. This supervision drives the encoder–decoder to learn a semantically discriminative and structurally coherent representation under fixed diffusion priors, which later serves as the distillation target.

2) Stage 2 (Knowledge Distillation via Decoder Feature Alignment): After the teacher converges, we train the student network [Fig. 2(b)]. The student shares the same VD encoder and decoder architecture as the teacher, but replaces the heavy diffusion sampler with a lightweight three-layer denoising U-Net [25] and integrates the proposed PSG module. The VD encoder and decoder are initialized from Stage 1; the encoder is frozen, and only the student diffusion U-Net and decoder are optimized.

In the diffusion branch, we follow a standard DDPM-style training objective. Given the encoder latent z_0, we sample a

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timestep t uniformly from $\{1, \ldots, 50\}$ and construct a noisy latent

$$z_t = \sqrt{\bar{\alpha}_t}\, z_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I) \tag{17}$$

where $\bar{\alpha}_t$ is the cumulative noise coefficient. The student diffusion U-Net ϵ_θ is then trained to predict the added noise

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{z_0,\varepsilon,t}\left[\|\varepsilon - \epsilon_\theta(z_t, t)\|_2^2\right]. \tag{18}$$

During refinement, we use a three-step DDIM sampler (on the same 50-step schedule) to denoise back to $t = 0$ and feed the refined latent to the decoder to obtain $\hat{Y}_S$. The student is supervised by

$$\mathcal{L}_{\text{task}} = \mathcal{L}_{\text{BCE}}\left(\hat{Y}_S, Y\right) + \mathcal{L}_{\text{Dice}}\left(\hat{Y}_S, Y\right) \tag{19}$$

where BCE and Dice jointly handle pixel-wise learning and foreground–background imbalance.

To further guide the student toward the teacher's representation space, we introduce a decoder FD loss. We select an intermediate feature map $f_T \in \mathbb{R}^{H' \times W' \times C'}$ from the fourth layer of the teacher decoder (chosen via ablation) and align it with the corresponding student feature f_S at the same resolution

$$\mathcal{L}_{\text{distill}} = \|f_S - f_T\|_2^2. \tag{20}$$

Since f_T and f_S are spatial feature maps, this loss enforces pixel-wise alignment in the decoder representation, enhancing structural awareness and boundary localization.

The overall objective in Stage 2 is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{diff}} + \alpha \cdot \mathcal{L}_{\text{task}} + \beta \cdot \mathcal{L}_{\text{distill}} \tag{21}$$

where α and β balance the contributions of each term. Unless otherwise stated, we set $\alpha = 1.0$ and $\beta = 0.5$; when the teacher is removed in ablation, β is set to zero. This two-stage training stabilizes optimization and enables the student to inherit the semantic and structural priors of the diffusion-augmented teacher with far fewer denoising steps, resulting in a compact and deployable ORSI-SOD model.

IV. Experiments

A. Experimental Setting

1) Implementation Details: All experiments were implemented in PyTorch and conducted on a single NVIDIA RTX A5000 GPU. Following common practice, images from ORSSD [13], EORSSD [14], and ORSI-4199 [15] were resized to a fixed resolution of 352×352. SGDDiff was optimized using the Adam optimizer with an initial learning rate of 6×10^{-5}, a batch size of 4, and a weight decay of 0.01. All models were trained for 32k iterations. The number of diffusion steps was empirically set to $T = 10$ for the teacher and $T = 3$ for the student, allowing the student to retain a lightweight diffusion refinement process while achieving a good balance between detection performance and computational efficiency.

2) Test Datasets: We evaluate the proposed SGDDiff on three widely used optical remote sensing SOD (ORSI-SOD) benchmark datasets.

1) *ORSSD [13]:* It consists of 800 remote sensing images, with 600 for training and 200 for testing. It features diverse object appearances and scene layouts.
2) *EORSSD [14]:* An extended and more complex version of ORSSD containing 2000 images (1400 for training and 600 for testing), with more challenging object distributions and higher scene variability.
3) *ORSI-4199 [15]:* A large-scale dataset containing 4199 images (2000 for training and 2199 for testing), characterized by complex backgrounds, cluttered scenes, and fine-grained object boundaries.

3) Compared Methods: We compare SGDDiff against 18 state-of-the-art methods, including both natural scene image SOD (NSI-SOD) and remote sensing image SOD (RSI-SOD) approaches: ASTTNet [33], ASNet [19], VST [37], LVNet [13], DAFNet [14], MCCNet [38], MJRBM [15], EMFINet [39], CorrNet [20], ERPNet-V [16], ACCoNet [40], SFANet [41], LiteSalNet [6], RAGRNet [17], AESINet [7], MEANet [18], MTPNet [8], and ORSIDiff [9].

4) Evaluation Metrics: We adopt four standard evaluation metrics widely used in SOD:

1) mean F-measure (F_β^{mean});
2) mean absolute error (MAE);
3) S-measure (S_α), with $\alpha = 0.5$;
4) mean E-measure (E_ξ^{mean}).

B. Comparison With State-of-the-Art Methods

1) Quantitative Results: We compare SGDDiff with 18 state-of-the-art methods on three ORSI-SOD benchmarks, i.e., EORSSD, ORSSD, and ORSI-4199. The quantitative results are summarized in Table I.

a) Results on EORSSD: On the EORSSD dataset, SGDDiff achieves superior performance on three of the four metrics.

1) *S-Measure (S_α):* SGDDiff obtains 0.9698, outperforming ORSIDiff (0.9482) and AESINet (0.9460) by clear margins.
2) *F-Measure (F_β^{mean}):* SGDDiff reaches 0.9201, slightly higher than AESINet (0.9183) and notably higher than ORSIDiff (0.8885).
3) *E-Measure (E_ξ^{mean}):* SGDDiff attains 0.9868, exceeding AESINet (0.9828) and ORSIDiff (0.9812).
4) *MAE:* SGDDiff records 0.0057, slightly higher than ORSIDiff (0.0042) and MTPNet (0.0048).

b) Results on ORSSD: On the ORSSD dataset, SGDDiff again achieves strong performance on three of the four metrics.

1) *S-Measure (S_α):* SGDDiff obtains 0.9621, outperforming ORSIDiff (0.9580) and MTPNet (0.9514).
2) *F-Measure (F_β^{mean}):* SGDDiff reaches 0.9286, slightly higher than ORSIDiff (0.9264) and higher than MTPNet (0.9179).
3) *E-Measure (E_ξ^{mean}):* SGDDiff attains 0.9875, exceeding ORSIDiff (0.9856) and RAGRNet (0.9818).

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TABLE I

QUANTITATIVE COMPARISON ON EORSSD, ORSSD, AND ORSI-4199. WE REPORT PARAMETER COUNT (PARAM, M), PER-STEP FLOPS (G), AND INFERENCE SPEED (FPS, IMAGES/S). THE BEST, SECOND-BEST, AND THIRD-BEST RESULTS ARE HIGHLIGHTED IN RED, BLUE, AND GREEN, RESPECTIVELY

Method	Year	Param	FLOPS	FPS	EORSSD				ORSSD				ORSI-4199			
					$S_\alpha \uparrow$	$F_\beta^{\text{mean}} \uparrow$	$E_\xi^{\text{mean}} \uparrow$	MAE $\downarrow$	$S_\alpha \uparrow$	$F_\beta^{\text{mean}} \uparrow$	$E_\xi^{\text{mean}} \uparrow$	MAE $\downarrow$	$S_\alpha \uparrow$	$F_\beta^{\text{mean}} \uparrow$	$E_\xi^{\text{mean}} \uparrow$	MAE $\downarrow$
VST [37]	2021	83.83	-	80	0.9008	0.7989	0.9312	0.0075	0.8811	0.7815	0.8926	0.0169	0.8592	0.8363	0.9210	0.0340
ASNet [19]	2024	65.4	20.01	61	0.9345	0.8770	0.9745	0.0055	0.9441	0.9072	0.9764	0.0081	-	-	-	-
LVNet [13]	2019	-	-	-	0.8644	0.7356	0.8826	0.0145	0.8815	0.7995	0.9259	0.0207	-	-	-	-
DAFNet [14]	2020	29.35	68.5	26	0.9185	0.7980	0.9382	0.0053	0.9118	0.8442	0.9537	0.0106	0.8451	0.8328	0.9119	0.0417
CorrNet [20]	2022	4.09	21.1	100	0.9289	0.8620	0.9646	0.0083	0.9380	0.9002	0.9746	0.0098	0.8623	0.8513	0.9206	0.0366
ASTTNet [33]	2023	-	-	13	0.9253	0.8297	0.9584	0.0059	0.9346	0.8811	0.9699	0.0094	0.8828	0.8673	0.9435	**0.0273**
MCCNet [38]	2021	67.65	112.8	95	0.9327	0.8604	0.9685	0.0066	0.9437	0.9054	0.9758	0.0087	-	-	-	-
MJRBM [15]	2021	63.28	80.56	19	0.9197	0.8239	0.9350	0.0099	0.9204	0.8566	0.9415	0.0163	0.8593	0.8309	0.9102	0.0374
EMFINet [39]	2022	107.26	480.9	25	0.9290	0.8486	0.9604	0.0084	0.9366	0.8856	0.9671	0.0109	0.8673	0.8624	0.9377	0.0315
ERPNet-V [16]	2022	56.48	87.2	17	0.9210	0.8632	0.9604	0.0089	0.9254	0.8974	0.9710	0.0135	0.8651	0.8671	0.9294	0.0373
ACCoNet [40]	2022	127.01	180.0	81	0.9290	0.8552	0.9653	0.0074	0.9437	0.8971	0.9754	0.0088	0.8775	0.8620	0.9342	0.0314
MEANet [18]	2024	3.27	9.62	42	0.9282	0.8678	0.9658	0.0070	0.9342	0.8934	0.9730	0.0098	-	-	-	-
SFANet [41]	2024	25.1	7.7	-	0.9349	0.8680	0.9726	0.0058	0.9453	0.9063	0.9789	0.0077	0.8761	0.8659	0.9385	0.0292
LiteSalNet [6]	2025	3.9	7.35	35	0.9273	0.8809	0.9634	0.0063	0.9373	0.9056	0.9700	0.0090	-	-	-	-
RAGRNet [17]	2024	35.6	17.8	18	0.9361	0.8770	0.9737	0.0057	0.9507	0.9157	0.9818	0.0066	0.8811	0.8763	0.9428	0.0284
AESINet [7]	2023	41.05	55.0	25	0.9460	0.9183	0.9828	0.0086	0.9358	0.8838	0.9751	0.0079	0.8702	0.8670	0.9403	0.0325
MTPNet [8]	2025	55.06	-	91	0.9394	0.8810	0.9775	0.0048	0.9514	0.9179	0.9833	0.0066	0.8838	0.8770	0.9451	**0.0269**
ORSIDiff [9]	2025	16.98	15.25	8	0.9482	0.8885	0.9812	**0.0042**	0.9580	0.9264	0.9856	**0.0051**	-	-	-	-
SGDDiff (Ours)	-	65.31	40	31	**0.9698**	**0.9201**	**0.9868**	0.0057	**0.9621**	**0.9286**	**0.9875**	0.0068	**0.8931**	**0.8791**	**0.9569**	0.0285

4) *MAE:* SGDDiff records 0.0068, slightly higher than ORSIDiff (0.0051) and also higher than MTPNet (0.0066) and RAGRNet (0.0066).

c) Results on ORSI-4199: On the ORSI-4199 dataset, SGDDiff remains competitive and achieves the best performance on three of the four metrics.

1) *S-Measure (S_α):* SGDDiff obtains 0.8931, outperforming MTPNet (0.8838) and ASTTNet (0.8828).
2) *F-Measure (F_β^{mean}):* SGDDiff reaches 0.8791, surpassing RAGRNet (0.8763) and ASTTNet (0.8673).
3) *E-Measure (E_ξ^{mean}):* SGDDiff attains 0.9569, exceeding MTPNet (0.9451) and ASTTNet (0.9435).
4) *MAE:* SGDDiff records 0.0285, slightly higher than MTPNet (0.0269) and ASTTNet (0.0273).

d) Parameter, FLOPs, and inference speed analysis: Beyond detection accuracy, we further evaluate model efficiency in terms of parameter count, FLOPs, and inference speed. SGDDiff contains 65.31 M parameters and requires 40 GFLOPs per step. Under the end-to-end evaluation protocol described above (RTX A5000, input size 352×352, batch size 4, and FP32 precision), SGDDiff achieves 31 FPS, and this reported speed already includes the complete three-step DDIM sampling process. Therefore, the proposed student model satisfies the real-time requirement (i.e., >30 FPS) under the reported setting. By comparison, the diffusion-based ORSIDiff runs at only 8 FPS, showing that SGDDiff achieves substantially higher throughput while also delivering better overall accuracy.

e) Precision–recall (PR) comparison: Fig. 5 shows the PR curves of SGDDiff against 13 representative methods on EORSSD, ORSSD, and ORSI-4199. SGDDiff consistently attains higher precision across a wide recall range, and exhibits slower performance degradation in the high-recall regime. On EORSSD and ORSSD, its curve stays closer to the upper left region, indicating both strong precision and recall. On the more challenging ORSI-4199 dataset, SGDDiff still maintains a clear advantage, suggesting better robustness under complex backgrounds and diverse object appearances.

f) Motivation-consistent analysis: These results are consistent with the motivation in Fig. 1. SGDDiff achieves consistent gains on F_β^{mean}, indicating improved foreground recall and reduced background confusion. The improvements on S_α and E_ξ^{mean} across all three datasets further suggest more structurally coherent predictions. Together, these gains demonstrate the robustness of SGDDiff in handling look-alike clutter, thin boundaries, and small objects. The slightly higher

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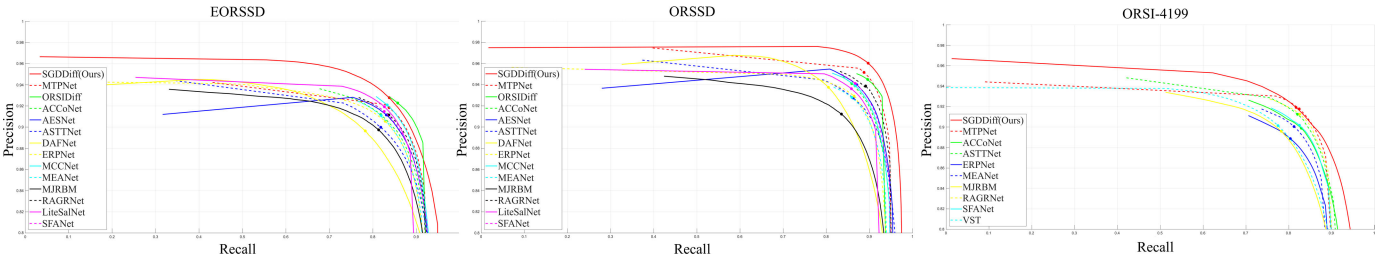


Fig. 5. Comparisons of PR curves on the EORSSD, ORSSD, and ORSI-4199 datasets.

MAE in some cases does not contradict this trend. MAE is a pixel-wise metric and is highly sensitive to local boundary deviations. In ORSI-SOD, lower MAE can sometimes be achieved by conservative predictions that shrink salient regions. In contrast, SGDDiff tends to preserve more complete object structures and reduce boundary overshrinking, which is beneficial for thin structures and small targets. Although this may introduce slight boundary expansion in some cases, it leads to more coherent salient objects overall. Therefore, SGDDiff shows stronger advantages on structure-aware metrics such as S_α, F_β^{mean}, and E_ξ^{mean} while remaining competitive on MAE. Overall, the accuracy gains together with real-time inference validate the intended accuracy–efficiency tradeoff for practical large-area ORSI-SOD deployment.

2) Qualitative Analysis: To complement the quantitative evaluation, we qualitatively compare SGDDiff with eight representative methods (MTPNet [8], AESINet [7], CorrNet [20], EMFINet [39], LiteSalNet [6], RAGRNet [17], SFANet [41], and ORSIDiff [9]) on EORSSD under three challenging conditions (Fig. 6). These baselines are selected due to their competitive performance in Table I. Overall, SGDDiff yields saliency maps with better structural integrity, clearer boundaries, and fewer background-induced errors.

a) Look-alike clutter (Rows 1–4): When foreground objects share similar appearance with surrounding background, many competing methods either miss the target or produce clutter-induced false responses. Benefiting from diffusion-style structural refinement distilled into the student and the training-time PSG, SGDDiff better suppresses distractors and preserves the complete object extent.

b) Thin/elongated structures (Rows 5–8): Linear objects such as roads and rivers are prone to fragmentation or over-smoothing. SGDDiff maintains continuity and recovers thin boundaries more reliably, producing more connected predictions with fewer breaks.

c) Small objects with large variation (Rows 9–12): Small targets with diverse shapes/scales (and sometimes spatial proximity) require both global context and local detail. Many baselines only detect a subset of objects or merge nearby instances into a single blob. In contrast, SGDDiff captures the object layout more faithfully and delineates boundaries more clearly.

These visual results align with the motivation in Fig. 1. SGDDiff mitigates the fragmentation of CNN/Transformer predictors and the clutter sensitivity of direct diffusion inference, enabling robust saliency detection in complex ORSI scenes.

TABLE II
ABLATION STUDIES ON EORSSD FOR THREE COMPONENTS: PSG, THE VD ENCODER (VD), AND FD

PSG	VD	FD	Param (M)	$S_\alpha \uparrow$	$F_\beta^{\text{mean}} \uparrow$	$E_\xi^{\text{mean}} \uparrow$	MAE ↓
			91.21	0.9241	0.8362	0.9101	0.0087
✓			105.71	0.9351	0.8653	0.9543	0.0075
✓	✓		128.32	0.9628	0.9017	0.9791	0.0063
✓	✓	✓	65.31	**0.9698**	**0.9201**	**0.9868**	**0.0057**

C. Ablation Experiments

We conduct ablation studies to isolate the contributions of the key designs in SGDDiff. Unless otherwise stated, all ablations are performed on EORSSD under the same training and evaluation protocol as the full model. Specifically, we incrementally add PSG, VD, and FD to quantify their individual and joint effects; evaluate the necessity of diffusion-based refinement during inference; analyze the VD encoder by ablating VGRNN (V) and DyT (D); study the kernel-size configuration in GDB; assess the effect of DyT on optimization stability; examine the impact of decoder-layer selection for feature alignment; and investigate the effect of DDIM sampling steps on accuracy and efficiency.

1) Ablation Study on Key Components: To evaluate the effectiveness of the proposed components, we conduct incremental ablation experiments on EORSSD by progressively introducing the PSG strategy, the VD encoder, and decoder FD. The quantitative results are summarized in Table II.

a) Row 1 (baseline: w/o PSG, VD, and FD): The baseline removes PSG and FD and replaces the VD encoder with the original Swin Transformer encoder while keeping all other settings unchanged. It achieves the lowest performance, with S_α = 0.9241, F_β^{mean} = 0.8362, E_ξ^{mean} = 0.9101, and MAE = 0.0087, indicating limited robustness in cluttered ORSI scenes without explicit semantic guidance or diffusion-aware knowledge transfer.

b) Row 2 (+ PSG): After introducing PSG, all metrics are consistently improved to S_α = 0.9351, F_β^{mean} = 0.8653, E_ξ^{mean} = 0.9543, and MAE= 0.0075. This confirms that label-driven pixel semantics can effectively reduce foreground–background ambiguity and improve regional consistency.

c) Row 3 (+ PSG + VD): Further incorporating the VD encoder significantly boosts performance to S_α = 0.9628, F_β^{mean} = 0.9017, E_ξ^{mean} = 0.9791, and MAE= 0.0063. This

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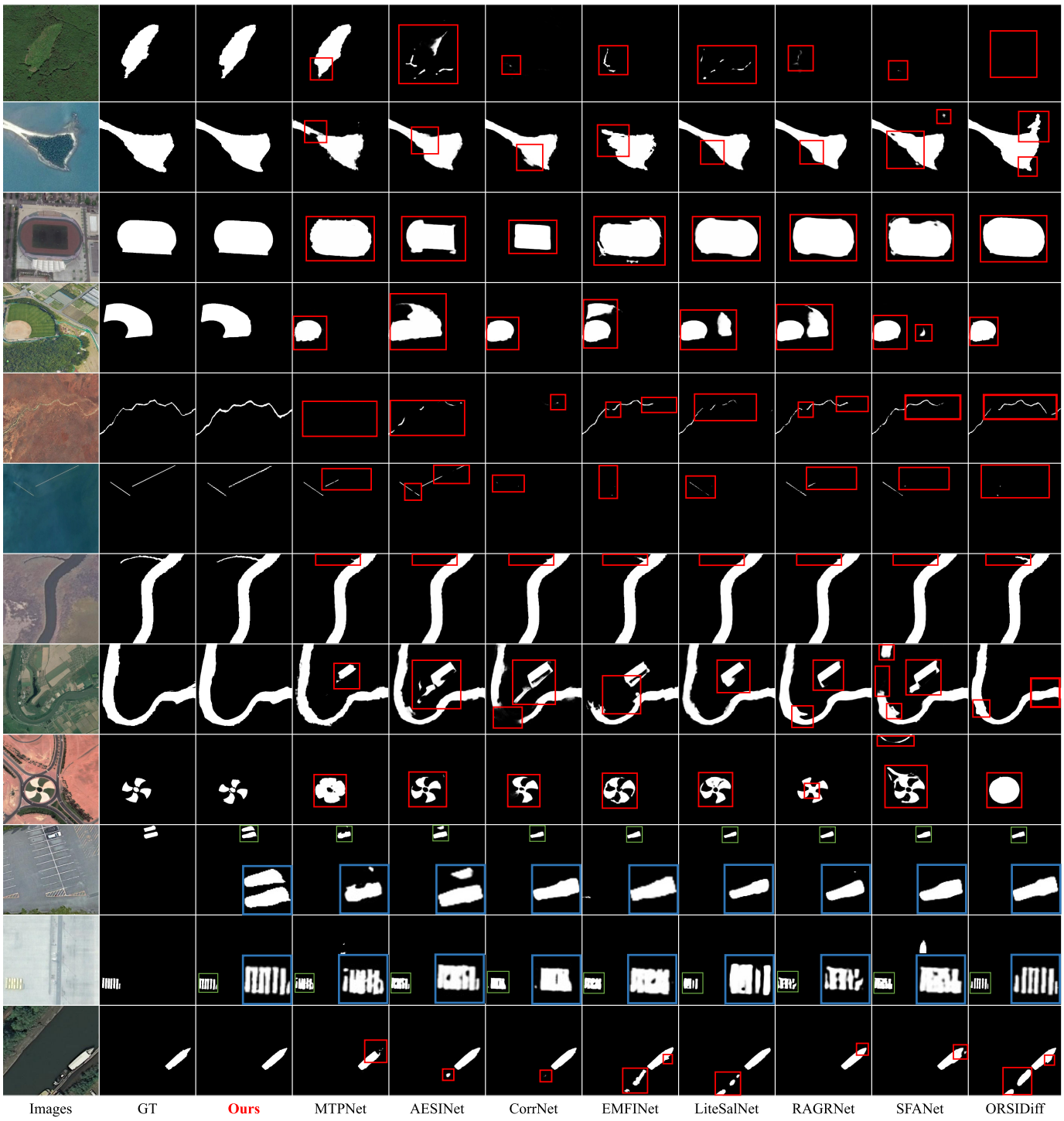


Fig. 6. Visual comparison of saliency predictions on EORSSD. "GT" denotes the ground truth. Rows 1–4: look-alike clutter. Rows 5–8: thin/elongated structures. Rows 9–12: small objects with large appearance variation. SGDDiff produces more complete targets with cleaner boundaries and fewer clutter-induced false responses (errors of other methods are highlighted with red boxes, regions of interest are marked with green boxes, and zoomed-in views are shown in blue).

demonstrates that the proposed encoder enhances long-range discrimination and local detail modeling, which is beneficial for accurately detecting small objects and preserving object boundaries.

d) Row 4 (full model: PSG + VD + FD): When FD is further introduced, the full model achieves the best performance, i.e., $S_\alpha = 0.9698$, $F_\beta^{\text{mean}} = 0.9201$, $E_\xi^{\text{mean}} = 0.9868$, and MAE = 0.0057. This verifies that decoder-level feature alignment effectively transfers diffusion-enhanced structural priors from the teacher to the student, resulting in more accurate and coherent saliency predictions.

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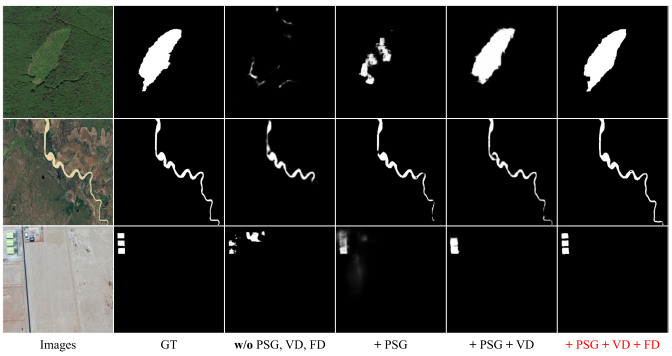


Fig. 7. Visual ablation results in three representative challenging scenarios. The qualitative comparisons show the progressive improvements introduced by different components. Specifically, PSG suppresses false positives in look-alike cluttered regions, the VD encoder enhances the detection of small-scale targets while improving both global discrimination and local detail preservation, and FD further improves structural completeness and continuity, which is particularly beneficial for thin salient objects.

TABLE III
ABLATION STUDY OF DIFFUSION-BASED INFERENCE ON EORSSD

Inference Strategy	$S_\alpha \uparrow$	$F_\beta^{\text{mean}} \uparrow$	$E_\xi^{\text{mean}} \uparrow$	MAE $\downarrow$
w/o Denoising Unet	0.9681	0.9188	0.9858	0.0061
w/ Denoising Unet (Ours)	**0.9698**	**0.9201**	**0.9868**	**0.0057**

TABLE IV
ABLATION STUDY OF THE VD ENCODER ON EORSSD. WE FURTHER VERIFY THE EFFECTS OF VGRNN (V) AND DYT (D), WHERE "W/O D" REPLACES DYT WITH A STANDARD NORMALIZATION LAYER

Model	$S_\alpha \uparrow$	$F_\beta^{\text{mean}} \uparrow$	$E_\xi^{\text{mean}} \uparrow$	MAE $\downarrow$
w/o V & D	0.9491	0.8992	0.9688	0.0065
w/o V	0.9523	0.9011	0.9710	0.0062
w/o D	0.9641	0.9082	0.9731	0.0060
Full VD	**0.9698**	**0.9201**	**0.9868**	**0.0057**

Fig. 7 further presents the visual ablation results. The qualitative comparisons indicate that PSG mainly suppresses look-alike clutter, the VD encoder improves both global discrimination and local detail preservation, and FD enhances structural completeness and spatial continuity. These findings are consistent with the quantitative results in Table II, confirming that the three components play complementary roles and jointly contribute to the overall performance improvement.

2) Effect of Diffusion-Based Refinement at Inference: We further evaluate whether keeping the student denoising U-Net at inference is beneficial. As shown in Table III, removing the denoiser degrades performance. This indicates that the few-step diffusion refinement improves structural integrity and boundary quality even at test time.

3) Effect of the VD Encoder: Table IV reports ablations on the VD encoder components, where **V** denotes the VGRNN refinement and **D** denotes DyT (replacing normalization; *w/o D* uses standard normalization). Compared with the Swin-only baseline (w/o V and D), adding either V or D improves performance while using both yields the best results. In particular, the full VD achieves the strongest overall accuracy, indicating complementary benefits from iterative refinement (V) and stable token calibration (D).

TABLE V
ABLATION ON GDB KERNEL-SIZE CONFIGURATIONS ON EORSSD

Kernel Size	$S_\alpha \uparrow$	$F_\beta^{\text{mean}} \uparrow$	$E_\xi^{\text{mean}} \uparrow$	MAE $\downarrow$
(1, 1, 1)	0.9686	0.9179	0.9856	0.0061
(3, 3, 3)	0.9695	0.9196	0.9865	0.0058
(5, 5, 5)	0.9596	0.9198	0.9866	0.0058
(1, 3, 5)	0.9694	0.9194	0.9865	0.0058
(3, 5, 3)	**0.9698**	**0.9201**	**0.9868**	**0.0057**

TABLE VI
ABLATION STUDY ON THE EFFECT OF FEATURE ALIGNMENT AT DIFFERENT DECODER LAYERS ON THE EORSSD DATASET

Alignment Layer	$S_\alpha \uparrow$	$F_\beta^{\text{mean}} \uparrow$	$E_\xi^{\text{mean}} \uparrow$	MAE $\downarrow$
Layer 2 (Early)	0.9591	0.9012	0.9722	0.0063
Layer 4 (Middle)	**0.9698**	**0.9201**	**0.9868**	**0.0057**
Layer 6 (Late)	0.9635	0.9146	0.9831	0.0061

4) Ablation on Kernel-Size Configuration in GDB: We vary the multibranch kernel sizes in the proposed GDB. As shown in Table V, the heterogeneous configuration (3, 5, 3) performs best among all candidates, suggesting that mixing medium and large receptive fields better captures multiscale cues in ORSIs while avoiding the limitations of uniformly small kernels or potential oversmoothing from uniformly large ones.

5) Effect of DyT on Optimization Stability: Fig. 8 compares training loss curves with and without DyT on EORSSD, ORSSD, and ORSI-4199. DyT consistently accelerates early stage loss reduction and produces smoother trajectories with smaller oscillations, indicating improved optimization stability under heterogeneous ORSI distributions. This stabilizing effect is particularly beneficial on ORSI-4199, where complex backgrounds typically induce stronger training noise.

6) Impact of Decoder-Layer Selection for Feature Alignment: We study where to apply feature alignment in the decoder. As shown in Table VI, aligning at the middle layer (Layer 4) yields the best overall results. Aligning too early (Layer 2) underperforms due to insufficient semantic abstraction, while aligning too late (Layer 6) is less effective likely because late features are more task-specific.

7) Effect of DDIM Sampling Steps on Accuracy and Efficiency: We further analyze the effect of the number of DDIM sampling steps on both detection accuracy and inference efficiency. As shown in Table VII, increasing the number of sampling steps consistently improves the detection performance. However, the improvement becomes marginal once the number of steps exceeds 3.

Meanwhile, the inference speed decreases steadily as more sampling steps are introduced. Although the four-step setting achieves slightly better numerical results than the three-step setting on some metrics, the gain is negligible compared with the additional computational cost. Therefore, the three-step configuration provides the best tradeoff between accuracy and

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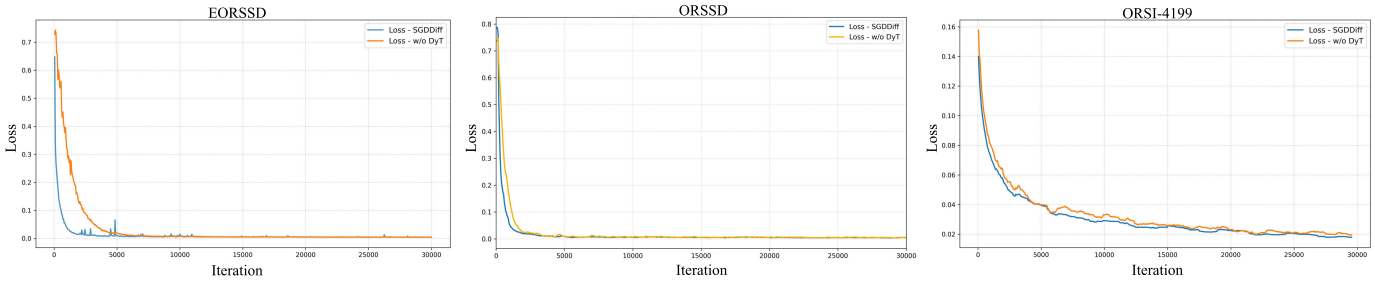


Fig. 8. Illustration of loss curves on the EORSSD, ORSSD, and ORSI-4199 datasets. With DyT, the model converges faster and exhibits smoother, more stable training curves.

TABLE VII

ABLATION STUDY ON THE NUMBER OF DDIM SAMPLING STEPS ON THE EORSSD DATASET. WE REPORT THE PER-STEP FLOPS, TOTAL FLOPS, END-TO-END INFERENCE SPEED (FPS, IMAGES/S), AND THE CORRESPONDING DETECTION PERFORMANCE UNDER DIFFERENT STEP SETTINGS

Steps	Per-step FLOPs	Total FLOPs	FPS	$S_\alpha \uparrow$	$F_\beta^{\text{mean}} \uparrow$	$E_\xi^{\text{mean}} \uparrow$	MAE $\downarrow$
1	40	51	36	0.9591	0.9031	0.9751	0.0062
2	40	91	33	0.9651	0.9174	0.9837	0.0059
3	40	131	**31**	**0.9698**	**0.9201**	**0.9868**	**0.0057**
4	40	171	28	0.9699	0.9203	0.9869	0.0057

efficiency, achieving strong detection performance while still maintaining real-time inference at 31 FPS.

It is also worth noting that the performance remains stable across adjacent step settings, which further demonstrates the robustness of the proposed framework. For a more practical evaluation of computational cost, in addition to the per-step FLOPs, we also report the total FLOPs and the end-to-end inference speed in FPS. All FPS results are measured on an NVIDIA RTX A5000 GPU with an input resolution of 352×352, a batch size of 4, and FP32 precision. Notably, the reported inference speed of the proposed student model includes the complete DDIM sampling process under each corresponding step setting.

V. CONCLUSION

In this article, we presented SGDDiff, a semantic-guided diffusion distillation network for ORSI-SOD. SGDDiff distills diffusion-style structural refinement into a compact student by performing *feature-space* diffusion distillation, where a diffusion teacher refines noised visual features into structurally coherent representations during training, and the student learns to approximate this refinement with only a few denoising steps at inference. To strengthen representation under strict efficiency constraints, we further introduce the VD encoder (VGRNN–DyT encoder), which integrates VGRNN-based iterative refinement and DyT-enhanced Swin-style context modeling within a single-stream hierarchy, as well as a PSG strategy that injects label-driven semantics along the denoising trajectory during training.

We adopt a two-stage learning scheme, including teacher representation construction under frozen diffusion priors and feature-aligned diffusion distillation via decoder feature alignment and semantic conditioning. This design enables the student to inherit diffusion-enhanced structural priors while remaining lightweight and deployable.

Extensive experiments on three ORSI-SOD benchmarks (EORSSD, ORSSD, and ORSI-4199) demonstrate that SGDDiff achieves state-of-the-art performance among 18 competing methods while meeting real-time inference requirements (31 FPS, i.e., >30 FPS), validating a favorable accuracy–efficiency tradeoff for large-area remote sensing deployment.

Although SGDDiff performs strongly overall, MAE can be slightly less competitive in some cases, suggesting that further improving pixel-level boundary precision remains valuable. In future work, we will explore stronger boundary-aware training objectives and more advanced diffusion distillation paradigms and extend the proposed framework to broader remote sensing dense prediction tasks.

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